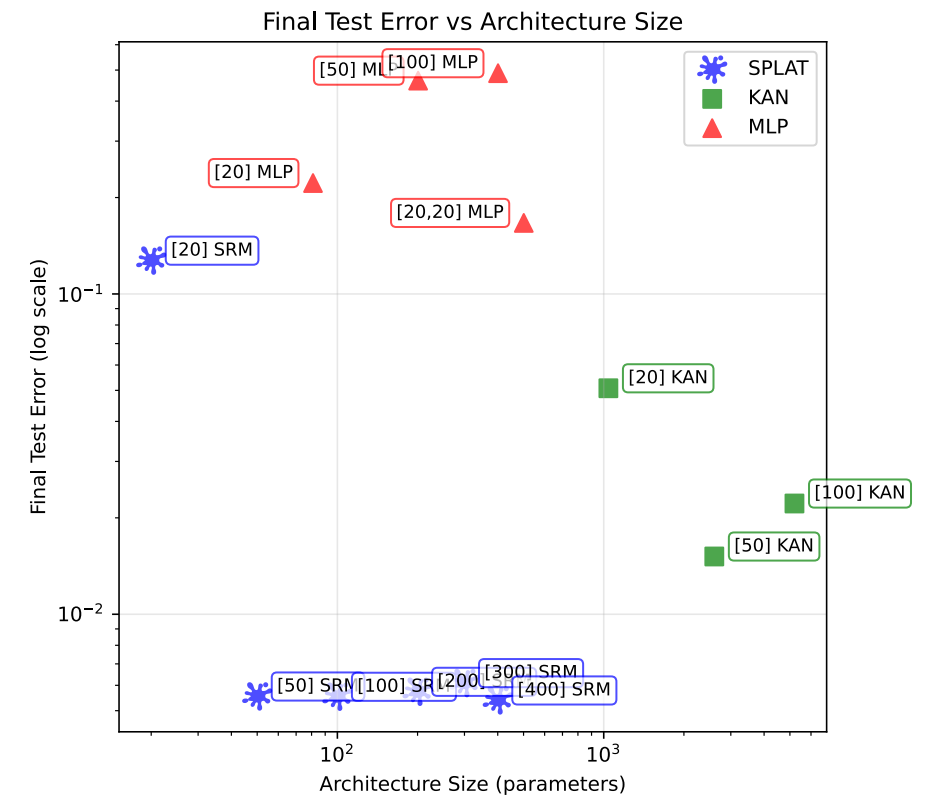
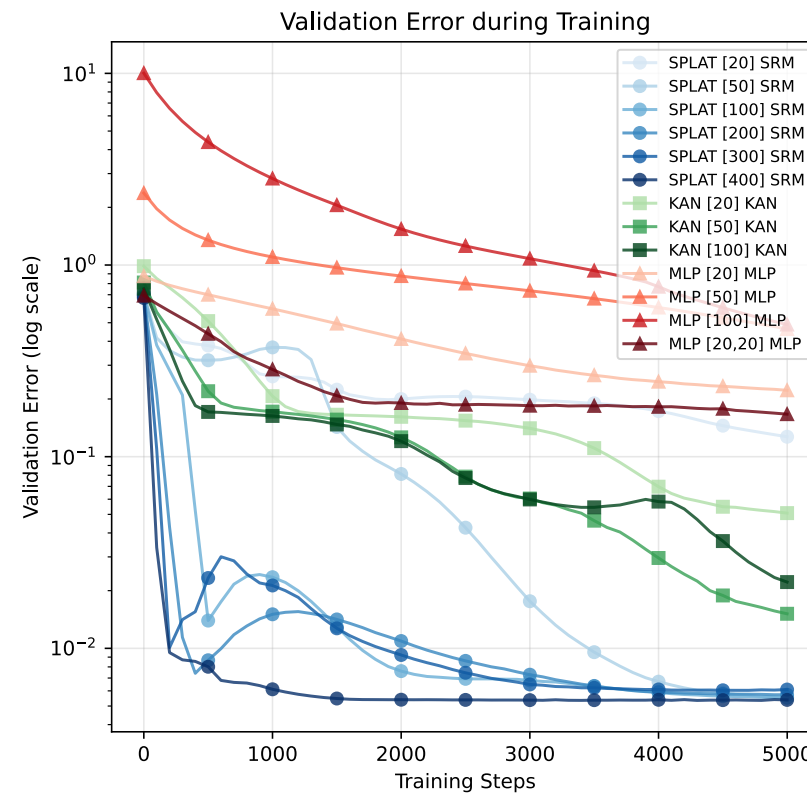
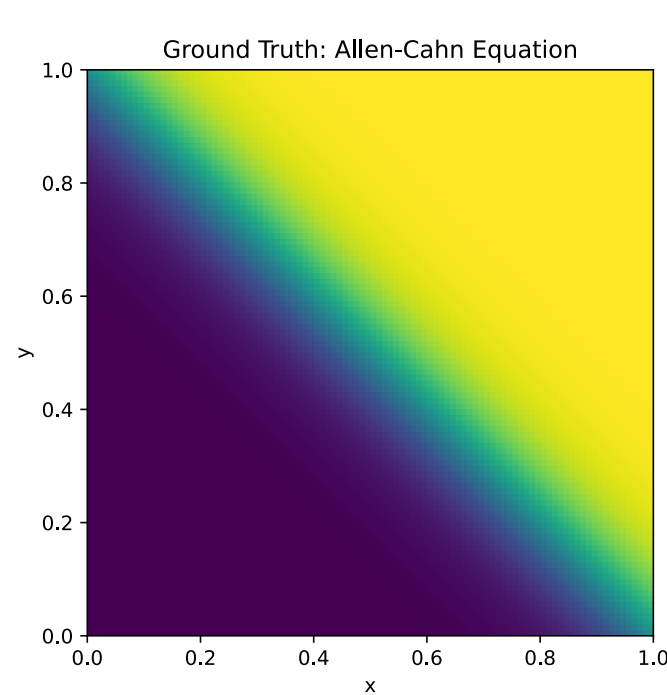


Experiments



PDE Fitting: $x_i \sim \mathcal{U}([0,1]^d)$, $i = 1 \dots 10^5$ and $z_j \sim \mathcal{U}(\partial[0,1]^d)$, $j = 1 \dots 5 \cdot 10^4$.

$$\mathcal{L}(u_\theta) = \frac{1}{n_{\text{int}}} \sum_{i=1}^{n_{\text{int}}} \|\mathcal{D}u_\theta(x_i) - f(x_i)\|_2^2 + \frac{1}{n_{\text{bdy}}} \sum_{j=1}^{n_{\text{bdy}}} \|u_\theta(z_j) - u^*(z_j)\|_2^2$$

$$\mathcal{D}u = 0.1\Delta u + u(1 - u^2)$$

Directions

Nonparametrics:

- Adaptive regression over inhomogeneous smoothness classes?
- Bias analysis, optimal splat approximation, comparisons to wavelets?

Optimization:

- Characterize approximate stationary points, escape times, metastable behavior.
- Establish quantitative convergence in WFR geometry.

Engineering:

- Use heuristics from graphics to develop better particle methods.
- Use principles from particle methods to improve splat-based modeling.